```oracle
SET SERVEROUTPUT ON;

DECLARE
    v_empno employees.employee_id%TYPE := &emp_id;
    v_salary employees.salary%TYPE;
    v_bonus  NUMBER := 0;
BEGIN
    SELECT salary INTO v_salary
    FROM employees
    WHERE employee_id = v_empno;

    IF v_salary IS NULL THEN
        v_bonus := 0;
    ELSIF v_salary < 1000 THEN
        v_bonus := v_salary * 0.10;
    ELSIF v_salary BETWEEN 1000 AND 1500 THEN
        v_bonus := v_salary * 0.15;
    ELSE
        v_bonus := v_salary * 0.20;
    END IF;

    DBMS_OUTPUT.PUT_LINE('Salary: ' || v_salary);
    DBMS_OUTPUT.PUT_LINE('Bonus: ' || v_bonus);

EXCEPTION
    WHEN NO_DATA_FOUND THEN
        DBMS_OUTPUT.PUT_LINE('Employee not found');
END;
/
```

Script Output

Task completed in 3.985 seconds

```
    DBMS_OUTPUT.PUT_LINE('Salary: ' || v_salary);
    DBMS_OUTPUT.PUT_LINE('Bonus: ' || v_bonus);

EXCEPTION
    WHEN NO_DATA_FOUND THEN
        DBMS_OUTPUT.PUT_LINE('Employee not found');
END;
Salary: 17000
Bonus: 3400


PL/SQL procedure successfully completed.
```

```oracle_plsql
SET SERVEROUTPUT ON;

DECLARE
    v_empno employees.employee_id%TYPE := &emp_id;
    v_salary employees.salary%TYPE;
    v_comm   employees.commission_pct%TYPE;
BEGIN
    SELECT salary, commission_pct
    INTO v_salary, v_comm
    FROM employees
    WHERE employee_id = v_empno;

    IF v_comm IS NULL THEN
        UPDATE employees
        SET salary = salary + 100
        WHERE employee_id = v_empno;

        DBMS_OUTPUT.PUT_LINE('Salary updated (commission was NULL)');
    ELSE
        DBMS_OUTPUT.PUT_LINE('Commission is not NULL');
    END IF;

EXCEPTION
    WHEN NO_DATA_FOUND THEN
        DBMS_OUTPUT.PUT_LINE('Employee not found');
END;
/
```

Script Output

Task completed in 4.039 seconds

```
        WHERE employee_id = v_empno;

        DBMS_OUTPUT.PUT_LINE('Salary updated (commission was NULL)');
    ELSE
        DBMS_OUTPUT.PUT_LINE('Commission is not NULL');
    END IF;

EXCEPTION
    WHEN NO_DATA_FOUND THEN
        DBMS_OUTPUT.PUT_LINE('Employee not found');
END;
Salary updated (commission was NULL)


PL/SQL procedure successfully completed.
```

```
SET SERVEROUTPUT ON;

DECLARE
    v_deptname departments.department_name%TYPE;
BEGIN
    SELECT department_name INTO v_deptname
    FROM departments
    WHERE department_id = 30;

    DBMS_OUTPUT.PUT_LINE('Department: ' || v_deptname);

EXCEPTION
    WHEN NO_DATA_FOUND THEN
        DBMS_OUTPUT.PUT_LINE('Department not found');
END;
/
```

Script Output

Task completed in 0.053 seconds

```
Department: Purchasing


PL/SQL procedure successfully completed.
```

```oracle_plsql
CREATE OR REPLACE PROCEDURE get_job(p_deptno NUMBER)
IS
BEGIN
    FOR rec IN (
        SELECT job_id FROM employees WHERE department_id = p_deptno
    )
    LOOP
        DBMS_OUTPUT.PUT_LINE('Job: ' || rec.job_id);
    END LOOP;
END;
/

BEGIN
    get_job(20);
END;
/
```

Script Output

Task completed in 0.049 seconds

```
Job: MK_MAN
Job: MK_REP


PL/SQL procedure successfully completed.
```

Worksheet | Query Builder

```
CREATE OR REPLACE PROCEDURE get_salary(p_deptno NUMBER)
IS
BEGIN
    FOR rec IN (
        SELECT salary FROM employees WHERE department_id = p_deptno
    )
    LOOP
        DBMS_OUTPUT.PUT_LINE('Salary: ' || rec.salary);
    END LOOP;
END;
/

BEGIN
    get_salary(20);
END;
/
```

Script Output

Task completed in 0.099 seconds

```
Procedure GET_SALARY compiled

Salary: 14000
Salary: 7000


PL/SQL procedure successfully completed.
```

Worksheet | Query Builder

```oracle_plsql
CREATE OR REPLACE PROCEDURE update_salary(p_empno NUMBER)
IS
BEGIN
    UPDATE employees
    SET salary = salary + (salary * 0.10)
    WHERE employee_id = p_empno;

    DBMS_OUTPUT.PUT_LINE('Salary increased by 10%');
END;
/

BEGIN
    update_salary(101);
END;
/
```

Script Output

Task completed in 0.279 seconds

```
Procedure UPDATE_SALARY compiled

Salary increased by 10%


PL/SQL procedure successfully completed.
```

```oracle_plsql
CREATE OR REPLACE PROCEDURE add_bonus(p_deptno NUMBER)
IS
BEGIN
    UPDATE employees
    SET salary = salary + 1000
    WHERE department_id = p_deptno
    AND salary > 5000;

    DBMS_OUTPUT.PUT_LINE('Bonus added to eligible employees');
END;
/

BEGIN
    add_bonus(20);
END;
/
```

Script Output

Task completed in 0.075 seconds

```
Procedure ADD_BONUS compiled

Bonus added to eligible employees


PL/SQL procedure successfully completed.
```

Worksheet | Query Builder

```sql
CREATE OR REPLACE VIEW vw_job_count AS
SELECT job_id, COUNT(*) AS total_employees
FROM employees
GROUP BY job_id;

CREATE OR REPLACE VIEW vw_emp_except_king AS
SELECT e.employee_id AS empno,
       e.first_name || ' ' || e.last_name AS empname,
       e.department_id AS deptno,
       d.department_name AS deptname
FROM employees e
JOIN departments d
ON e.department_id = d.department_id
WHERE UPPER(e.last_name) <> 'KING';

CREATE OR REPLACE VIEW vw_emp_details AS
SELECT e.employee_id AS empno,
       e.first_name || ' ' || e.last_name AS empname,
       e.department_id AS deptno,
       d.department_name AS deptname
FROM employees e
JOIN departments d
ON e.department_id = d.department_id;
```

Script Output

Task completed in 0.083 seconds

```
View VW_JOB_COUNT created.


View VW_EMP_EXCEPT_KING created.


View VW_EMP_DETAILS created.
```

Worksheet | Query Builder

```
SET SERVEROUTPUT ON;

DECLARE
    a NUMBER := &num1;
    b NUMBER := &num2;
    sum_result NUMBER;
BEGIN
    sum_result := a + b;
    DBMS_OUTPUT.PUT_LINE('Sum = ' || sum_result);
END;
/
```

Script Output

Task completed in 16.033 seconds

```
old:DECLARE
    a NUMBER := &num1;
    b NUMBER := &num2;
    sum_result NUMBER;
BEGIN
    sum_result := a + b;
    DBMS_OUTPUT.PUT_LINE('Sum = ' || sum_result);
END;

new:DECLARE
    a NUMBER := 1510;
    b NUMBER := 3008;
    sum_result NUMBER;
BEGIN
    sum_result := a + b;
    DBMS_OUTPUT.PUT_LINE('Sum = ' || sum_result);
END;

PL/SQL procedure successfully completed.
```

```
SET SERVEROUTPUT ON;

DECLARE
    low NUMBER := &low;
    high NUMBER := &high;
    total NUMBER := 0;
BEGIN
    FOR i IN low..high LOOP
        total := total + i;
    END LOOP;

    DBMS_OUTPUT.PUT_LINE('Sum = ' || total);
END;
/
```

Script Output

Task completed in 6.394 seconds

```
old:DECLARE
    low NUMBER := &low;
    high NUMBER := &high;
    total NUMBER := 0;
BEGIN
    FOR i IN low..high LOOP
        total := total + i;
    END LOOP;

    DBMS_OUTPUT.PUT_LINE('Sum = ' || total);
END;

new:DECLARE
    low NUMBER := 2;
    high NUMBER := 43;
    total NUMBER := 0;
BEGIN
    FOR i IN low..high LOOP
        total := total + i;
    END LOOP;

    DBMS_OUTPUT.PUT_LINE('Sum = ' || total);
END;

PL/SQL procedure successfully completed.
```

Worksheet | Query Builder

```oracle_plsql
SET SERVEROUTPUT ON;

DECLARE
    v_empno employees.employee_id%TYPE := &emp_id;
    v_name  VARCHAR2(100);
    v_hire  DATE;
    v_dept  VARCHAR2(100);
BEGIN
    SELECT e.first_name || ' ' || e.last_name,
           e.hire_date,
           d.department_name
    INTO v_name, v_hire, v_dept
    FROM employees e
    JOIN departments d
    ON e.department_id = d.department_id
    WHERE e.employee_id = v_empno;

    DBMS_OUTPUT.PUT_LINE('Name: ' || v_name);
    DBMS_OUTPUT.PUT_LINE('Hire Date: ' || v_hire);
    DBMS_OUTPUT.PUT_LINE('Department: ' || v_dept);

EXCEPTION
    WHEN NO_DATA_FOUND THEN
        DBMS_OUTPUT.PUT_LINE('Employee not found');
END;
/
```

Script Output

Task completed in 3.792 seconds

```
    FROM employees e
    JOIN departments d
    ON e.department_id = d.department_id
    WHERE e.employee_id = v_empno;

    DBMS_OUTPUT.PUT_LINE('Name: ' || v_name);
    DBMS_OUTPUT.PUT_LINE('Hire Date: ' || v_hire);
    DBMS_OUTPUT.PUT_LINE('Department: ' || v_dept);

EXCEPTION
    WHEN NO_DATA_FOUND THEN
        DBMS_OUTPUT.PUT_LINE('Employee not found');
END;

PL/SQL procedure successfully completed.


Substitution cancelled
```

Worksheet | Query Builder

```
SET SERVEROUTPUT ON;

DECLARE
    num NUMBER := &num;
    original NUMBER;
    rev NUMBER := 0;
    rem NUMBER;
BEGIN
    original := num;

    WHILE num > 0 LOOP
        rem := MOD(num, 10);
        rev := rev * 10 + rem;
        num := FLOOR(num / 10);
    END LOOP;

    IF original = rev THEN
        DBMS_OUTPUT.PUT_LINE('Palindrome');
    ELSE
        DBMS_OUTPUT.PUT_LINE('Not Palindrome');
    END IF;
END;
/
```

Script Output

Task completed in 3.728 seconds

```
    WHILE num > 0 LOOP
        rem := MOD(num, 10);
        rev := rev * 10 + rem;
        num := FLOOR(num / 10);
    END LOOP;

    IF original = rev THEN
        DBMS_OUTPUT.PUT_LINE('Palindrome');
    ELSE
        DBMS_OUTPUT.PUT_LINE('Not Palindrome');
    END IF;
END;
Not Palindrome


PL/SQL procedure successfully completed.
```

Worksheet | Query Builder

```
SET SERVEROUTPUT ON;

DECLARE
    v_empno NUMBER := &empno;
    v_fname VARCHAR2(50) := '&fname';
    v_lname VARCHAR2(50) := '&lname';
    v_deptno NUMBER := &deptno;
    v_dname VARCHAR2(50) := '&deptname';
BEGIN
    -- Insert into department (if not exists)
    INSERT INTO departments (department_id, department_name)
    VALUES (v_deptno, v_dname);

    -- Insert into employees
    INSERT INTO employees (employee_id, first_name, last_name, department_id, hire_date)
    VALUES (v_empno, v_fname, v_lname, v_deptno, SYSDATE);

    DBMS_OUTPUT.PUT_LINE('Records inserted');

EXCEPTION
    WHEN DUP_VAL_ON_INDEX THEN
        DBMS_OUTPUT.PUT_LINE('Duplicate entry detected');
    WHEN OTHERS THEN
        DBMS_OUTPUT.PUT_LINE('Error occurred');
END;
/
```

Script Output

Task completed in 26.364 seconds

```
    -- Insert into employees
    INSERT INTO employees (employee_id, first_name, last_name, department_id, hire_date)
    VALUES (v_empno, v_fname, v_lname, v_deptno, SYSDATE);

    DBMS_OUTPUT.PUT_LINE('Records inserted');

EXCEPTION
    WHEN DUP_VAL_ON_INDEX THEN
        DBMS_OUTPUT.PUT_LINE('Duplicate entry detected');
    WHEN OTHERS THEN
        DBMS_OUTPUT.PUT_LINE('Error occurred');
END;
Error occurred


PL/SQL procedure successfully completed.
```

```oracle
SET SERVEROUTPUT ON;

DECLARE
    v_manager NUMBER;
    v_name VARCHAR2(100);
    v_salary NUMBER;
BEGIN
    -- Start from employee 90
    SELECT manager_id INTO v_manager
    FROM employees
    WHERE employee_id = 90;

    WHILE v_manager IS NOT NULL LOOP
        SELECT first_name || ' ' || last_name, salary, manager_id
        INTO v_name, v_salary, v_manager
        FROM employees
        WHERE employee_id = v_manager;

        IF v_salary > 2500 THEN
            DBMS_OUTPUT.PUT_LINE('Employee: ' || v_name);
            DBMS_OUTPUT.PUT_LINE('Salary: ' || v_salary);
            EXIT;
        END IF;
    END LOOP;

EXCEPTION
    WHEN NO_DATA_FOUND THEN
        DBMS_OUTPUT.PUT_LINE('No such employee');
END;
/
```

Script Output

Task completed in 0.07 seconds

```
No such employee


PL/SQL procedure successfully completed.
```

```
SET SERVEROUTPUT ON;

DECLARE
    total NUMBER := 0;
BEGIN
    FOR i IN 1..100 LOOP
        total := total + i;
    END LOOP;

    DBMS_OUTPUT.PUT_LINE('Sum = ' || total);
END;
/
```

Script Output

Task completed in 0.105 seconds

```
Sum = 5050


PL/SQL procedure successfully completed.
```